Computer Graphics Project Proposal

Escape From Tiamat

Project Name: Escape From Tiamat

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Brief Introduction

Escape From Tiamat is a first-person escape-puzzle game developed using Three.js and WebGL. The player is trapped inside a damaged submarine and must solve a series of interactive puzzles to escape. The game features flashlight-based lighting, mouse-picking interaction, and animated objects within a fully 3D environment.

Detailed Description

Scene & Atmosphere

The game takes place in a single room inside the submarine Tiamat. The room has four walls; one wall contains the exit door, while each of the remaining three walls hosts a distinct puzzle. A table is placed at the center of the room. The environment is dominated by metallic surfaces, pipes, and industrial equipment. All major surfaces will be rendered with distinct textures and normal maps to convey depth and material detail, with specular highlights visible on metallic objects. The scene contains at least three object types with different morphologies and textures, satisfying the minimum requirements.

Lighting

The primary light source is a flashlight carried by the player, implemented as a spot light that moves with the camera. The player can increase or decrease the brightness of the flashlight, satisfying the minimum requirements for a movable light source with adjustable intensity. Additional wall-mounted lights are placed around the room as point lights with low intensity, ensuring that textures, normal maps, and specular highlights remain visible throughout the scene. All lighting is calculated using the Phong shading model, which simulates ambient, diffuse, and specular components for realistic surface rendering.

Object & Texture Design

The scene contains several object types with distinct morphologies, including a door, valves, a pressure gauge, a control panel, pipes, and a table. Each object type features a unique texture, with materials ranging from rusted metal and painted steel to plastic and wood. Normal maps are applied to major surfaces to simulate surface detail and depth without additional geometry. Specular highlights are configured for metallic objects to reflect the flashlight realistically. The scene satisfies the minimum requirements of at least three object types with different morphologies and at least three different textures.

Interaction

The player navigates the scene in first-person view using keyboard and mouse input, with full movement across three axes and free rotation in three dimensions, satisfying the minimum camera requirements. Object interaction is handled through raycasting; when the player's cursor hovers over an interactable object, the object is highlighted and its name is displayed on screen. The player can interact with objects by clicking. At least three objects in the scene are fully controllable by the player, including the valves, the pressure gauge, and the control panel, satisfying the minimum requirements.

Gameplay & Puzzles

The player is trapped inside the submarine Tiamat and must solve three puzzles to unlock the exit door and escape. Each puzzle is located on a separate wall of the room. The puzzles involve operating valves, adjusting a pressure gauge, and replicating a light sequence on a control panel. Once all three puzzles are solved, the exit door unlocks and the player escapes.

Additional Features

The project includes several features beyond the minimum requirements. Interactable objects such as valves and the exit door are animated, providing visual feedback when the player interacts with them. Hovering over an interactable object highlights it and displays its name on screen. Normal maps are applied to major surfaces to enhance visual depth. Specular highlights on metallic surfaces and the Phong shading model are implemented to achieve realistic lighting.

References

- Three.js Documentation — https://threejs.org/docs
- Angel, E. & Shreiner, D. — Interactive Computer Graphics